

R&I Interns Day



Remote Sensing for Sustainability

Name of intern: Parv Jain

College: Indian Institute Of Technology Tirupati

Mentor: Mr. Shailesh Deshpande Sir

Research Area: Learning Chromatic Primaries for Hyperspectral Artificial Vision

Period : 20th May 2026 – 18 Dec 2026



Learning & Implementation Journey

📖 Patent Study

Studied the patented framework for Hyperspectral Artificial Vision.

🌈 Hyperspectral Imaging

Understood spectral signatures and material discrimination.

🔍 SRFs & Primaries

Explored SRF learning and optimal chromatic primaries.

📖 Literature Review

Analyzed existing methods and prior-art research.

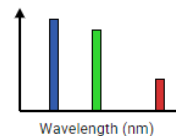
🧠 KAN Exploration

Studied KAN architecture and implementation.

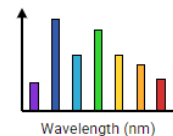
🚀 Ongoing Work

Implementing a KAN-based framework for learning chromatic primaries from hyperspectral data.

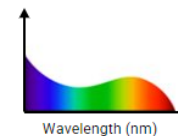
RGB
3 separated bands



MULTISPECTRAL
N separated bands



HYPERSPPECTRAL
Continuous Spectrum



Hyperspectral imaging captures continuous spectral information across multiple wavelength bands, providing richer material information than conventional RGB imaging.

What was the proposed Solution?

Investigate the use of **Kolmogorov-Arnold Networks (KAN)** for learning spectral representations from hyperspectral data.

- Explore its applicability for chromatic primary and SRF learning.
- Evaluate the learned representations for artificial vision tasks.

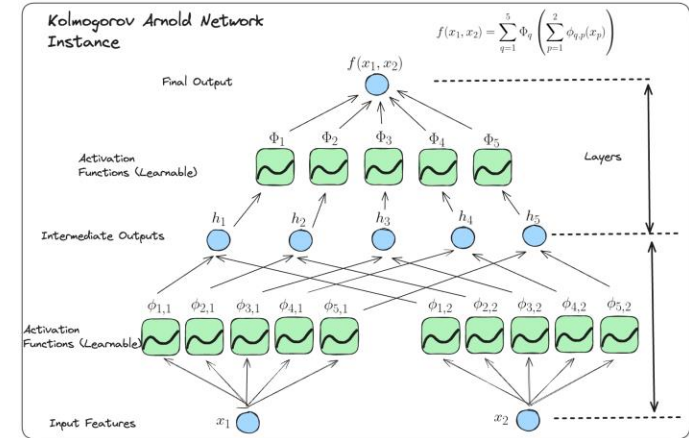
Tools/Technologies used:

- Python
- PyTorch
- KAN Framework
- NumPy, Pandas
- Matplotlib
- Jupyter Notebook
- Hyperspectral Datasets



Planned Methodology

- Hyperspectral Data Analysis
- Spectral Response Function (SRF) Learning
- Chromatic Primary Learning
- Deep Learning / KAN-based Modeling
- Classification & Performance Evaluation
- Comparative Analysis with Existing Approaches



Outcome & Learnings

Performance Comparison MLP vs KAN

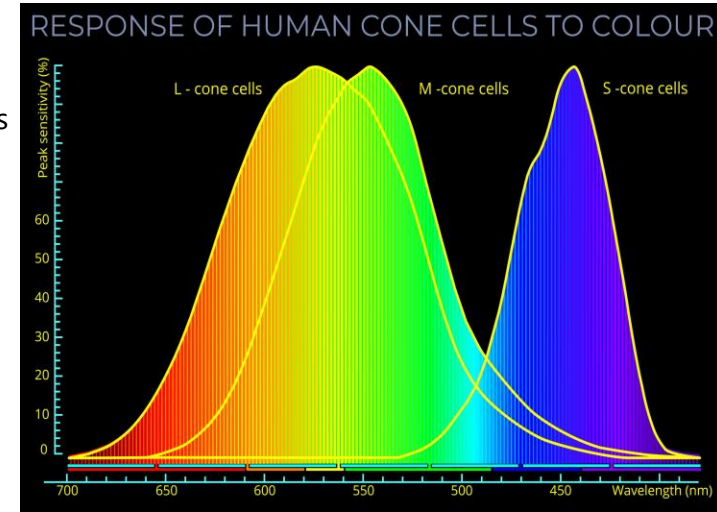
Model	OA (%)	AA (%)	Kappa
1D-KAN	97.66	96.64	0.9733
MLP-ReLU	95.96	94.09	0.9540
MLP-Sigmoid	95.55	93.22	0.9493
MLP-ReLU+Sigmoid	95.87	93.49	0.9529

Key learnings and skills gained

- Understanding of hyperspectral imaging and artificial vision concepts.
- Study of SRFs, chromatic primaries, and optimal primitives.
- Hands-on experience with KAN implementation using PyTorch.
- Experience in training, evaluating, and comparing deep learning models on hyperspectral datasets.

Upcoming Work

- Implement the **KAN-based framework** for learning Spectral Response Functions (SRFs) from hyperspectral data.
- Study whether learned **tri-chromatic SRFs** resemble conventional CIE/XIE color matching functions.
- Analyze the effect of varying the **number of chromatic primaries** on classification performance.
- Investigate whether **more than three primaries** provide better material discrimination in hyperspectral scenes.
- Compare KAN-based results with the existing approach in terms of:
 - Accuracy
 - Convergence
 - Computational Complexity
- Evaluate learned SRFs using visualization and similarity-based metrics.



Most memorable moments of the internship

Internship Journey Highlights

22nd May 2025 – First Day at TCS R&I

- First time entering the TCS Research & Innovation building.
- New environment, new colleagues, and the start of an exciting learning journey.
- My first experience in a corporate research setting.

10th June 2025 – Meeting My Mentor & Team

- First in-person visit to the QUADRA office.
- Met my mentor and team face-to-face after online discussions.
- A memorable moment that strengthened my connection to the project and team

"Every great journey starts with a first step—mine started at TCS, and every day since has been a new opportunity to learn and grow."



Mentor–Intern Conversations



Shailesh Deshpande Sir

Me: "Why do we use 3 chromatic primaries?"

Mentor: "Because humans do."

Me: "And machines?"

Mentor: "That's exactly what we're trying to find out."

Key Insight:

Can artificial vision systems learn chromatic primaries that are better suited for machines than those evolved for human vision?



Parv Jain



Thank You

